

A Comprehensive Study of ML Techniques for Quantum Error Mitigation

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Introduction

- ▶ Quantum errors limit the potential of quantum computing.
- ▶ Existing QEM methods are effective but costly.
- ▶ ML offers efficient alternatives for error mitigation.
- ▶ Few papers on QEM were reviewed and are cited in the references.

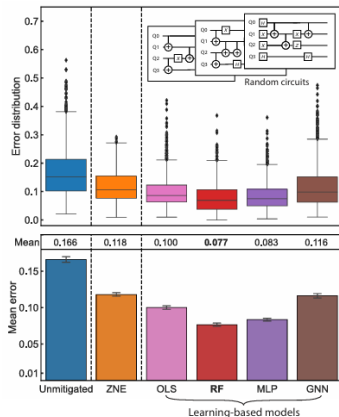


Figure: Past Workings

Project Overview

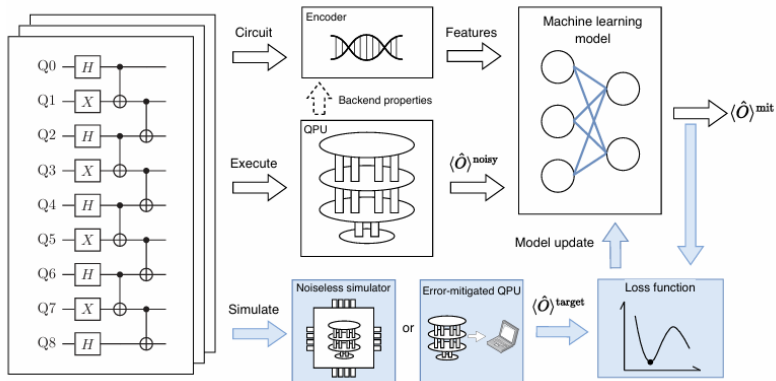


Figure: Overview of Project (Image source: [1])

Kolmogorov Arnold Networks (KANs)

- ▶ KANs provide a new approach to function approximation.
- ▶ Potential for better error mitigation in quantum computing.

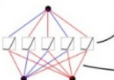
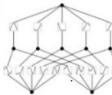
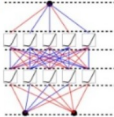
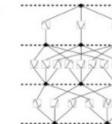
Model	Multi-Layer Perceptron (MLP)	Kolmogorov-Arnold Network (KAN)
Theorem	Universal Approximation Theorem	Kolmogorov-Arnold Representation Theorem
Formula (Shallow)	$f(\mathbf{x}) \approx \sum_{i=1}^{N(\epsilon)} a_i \sigma(\mathbf{w}_i \cdot \mathbf{x} + b_i)$	$f(\mathbf{x}) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right)$
Model (Shallow)	(a)  fixed activation functions on nodes learnable weights on edges	(b)  learnable activation functions on edges sum operation on nodes
Formula (Deep)	$\text{MLP}(\mathbf{x}) = (\mathbf{W}_3 \circ \sigma_2 \circ \mathbf{W}_2 \circ \sigma_1 \circ \mathbf{W}_1)(\mathbf{x})$	$\text{KAN}(\mathbf{x}) = (\Phi_3 \circ \Phi_2 \circ \Phi_1)(\mathbf{x})$
Model (Deep)	(c)  MLP(x) $\mathbf{W}_3$ $\mathbf{W}_2$ σ_2 $\mathbf{W}_1$ σ_1 $\mathbf{x}$ nonlinear, fixed linear, learnable	(d)  KAN(x) Φ_3 Φ_2 Φ_1 $\mathbf{x}$ nonlinear, learnable

Figure: Rough illustration of KAN network (Image source: [4])

Graph Attention Neural Networks (GATs)

- ▶ Investigating alternative neural network architectures.
- ▶ GATs enable more context-aware quantum error mitigation.
- ▶ Since quantum circuits are stored as graph form, we are thinking of trying out this type of network to understand which part of the circuit is important.

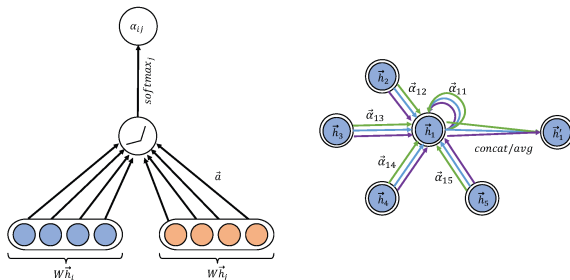


Figure: Rough illustration of GAT network (Image source: [?])

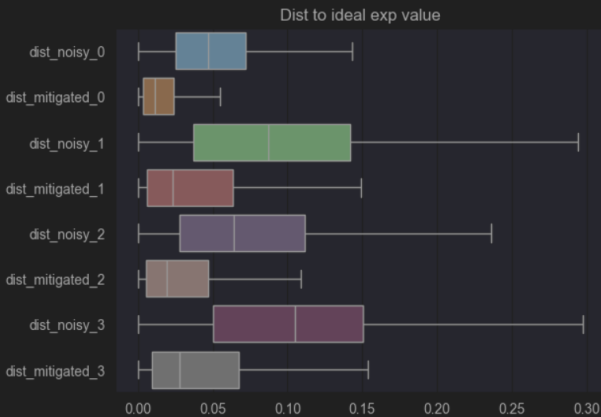
Project Plan

- ▶ Phase 1: Understanding Data Generation (Feb 10 - Feb 24)
- ▶ Phase 2: KAN Architecture Exploration (Feb 25 - Mar 10)
- ▶ Phase 3: Implementation (Mar 11 - Mar 25)
- ▶ Phase 4: Model Exploration (Mar 26 - Apr 9)
- ▶ Phase 5: Explainability Study (Apr 10 - Apr 24)
- ▶ Phase 6: Final Implementation (Apr 25 - May 9)

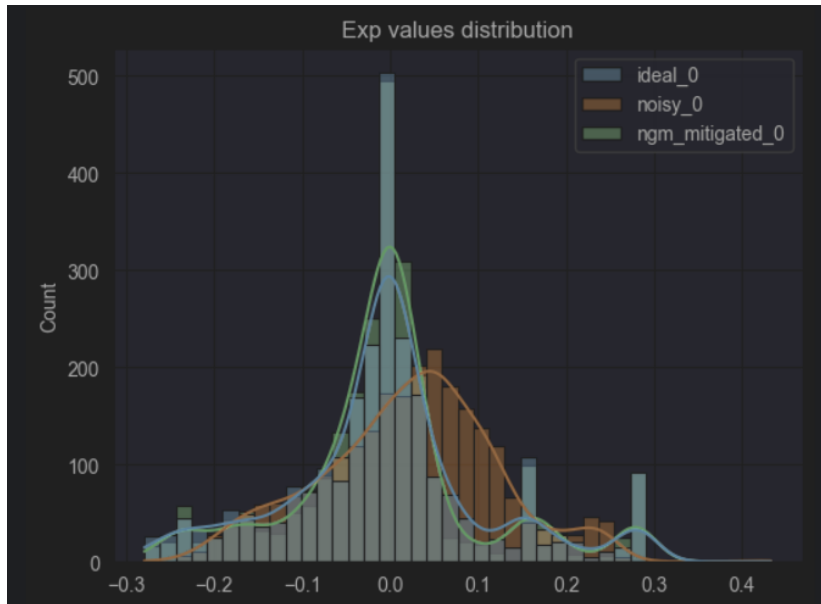
We are still in the process of understanding the data-generating process and exploring even more ML models that we might be able to implement.

Random Forest 4q ising

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Random Forest 4q ising

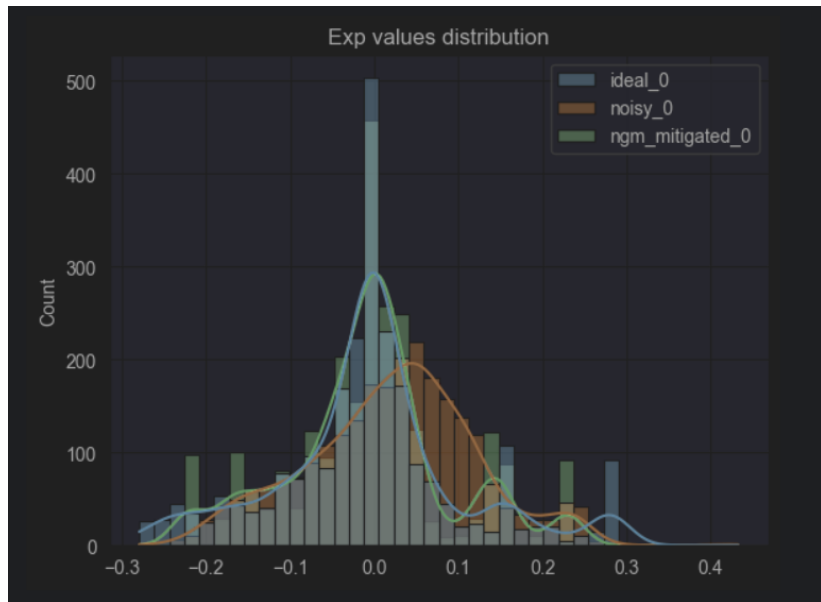


KAN 4q ising

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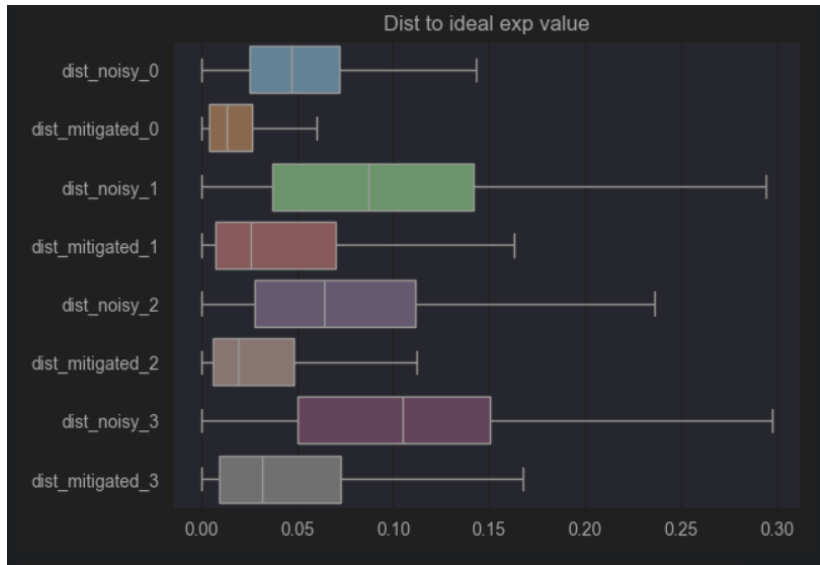
KAN 4q ising



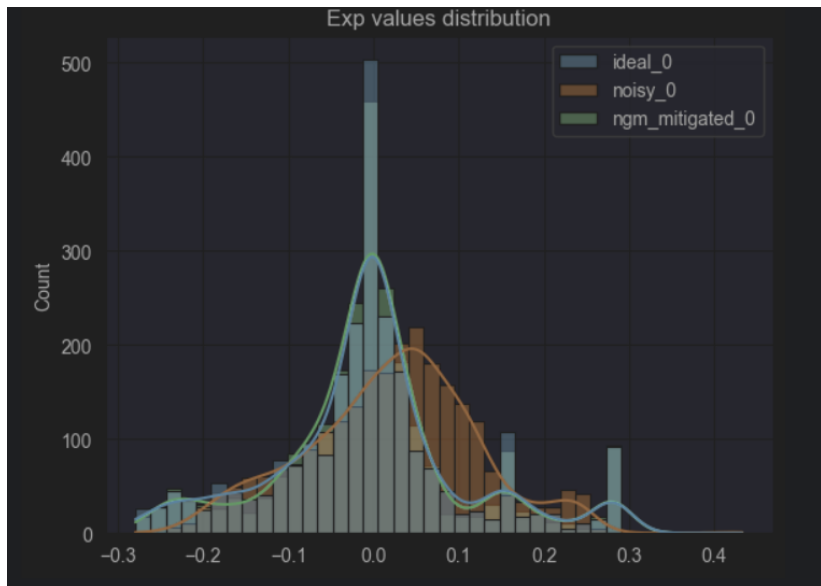
XGBoost 4q ising

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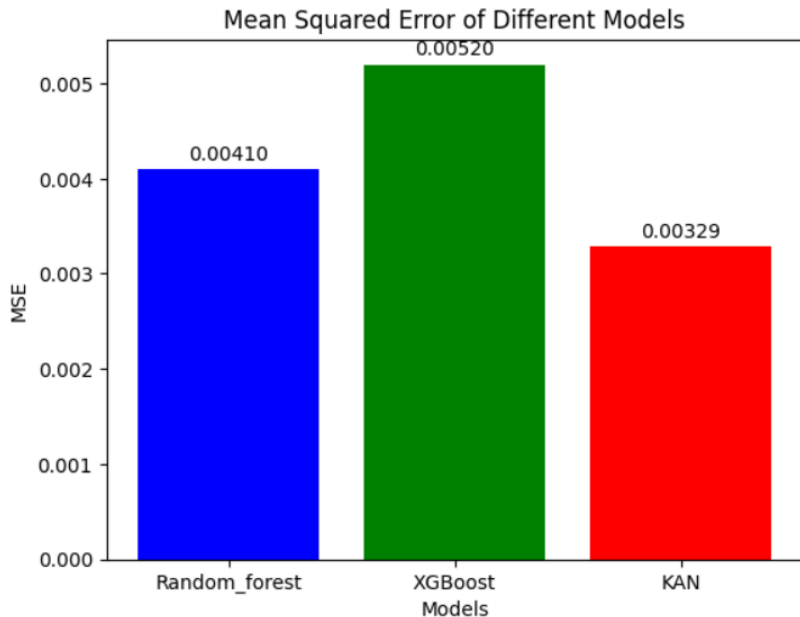
XGBoost 4q ising



XGBoost 4q ising



Result (Graph dataset)



References

-  Haoran Liao et al., "Machine Learning for Practical Quantum Error Mitigation," IBM Quantum, 2025.
-  Xiao-Yue Xu et al., "Physics-inspired Machine Learning for Quantum Error Mitigation," Henan Key Laboratory, 2025.
-  Asmar Muqet et al., "A ML-Based Error Mitigation Approach for IBM Quantum Computers," Simula Research, 2024.
-  Ziming Liu et al., "KAN: Kolmogorov–Arnold Networks," MIT, 2024.